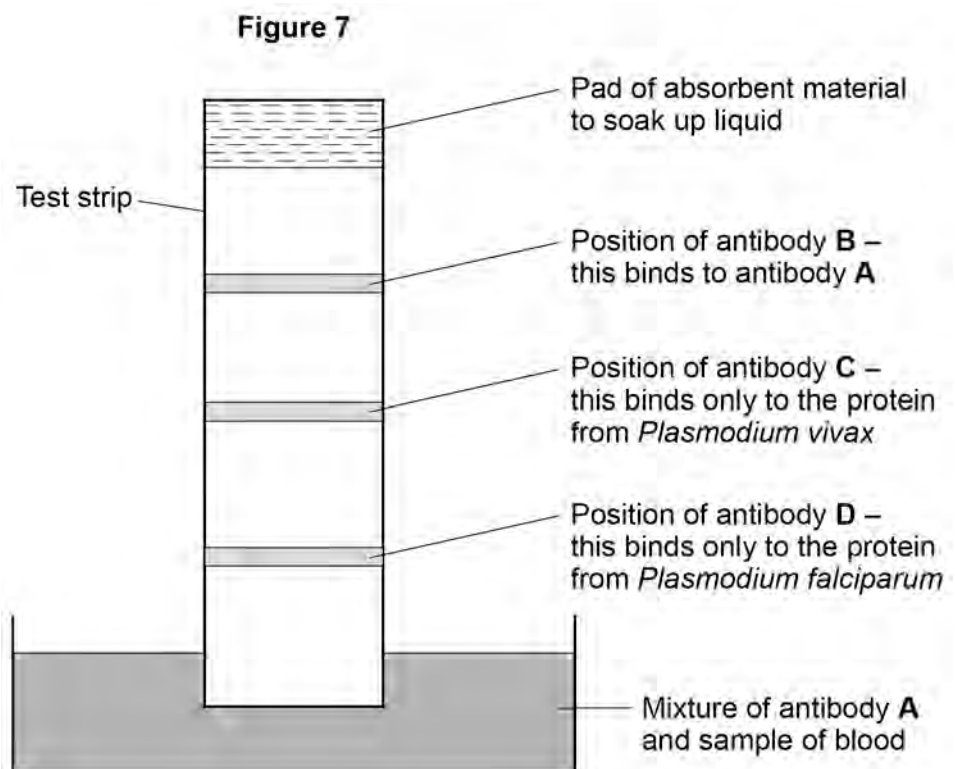


8

Malaria is a disease caused by parasites belonging to the genus *Plasmodium*. Two species that cause malaria are *Plasmodium falciparum* and *Plasmodium vivax*.

A test strip that uses monoclonal antibodies can be used to determine whether a person is infected by *Plasmodium*. It can also be used to find which species of *Plasmodium* they are infected by.

- A sample of a person's blood is mixed with a solution containing an antibody, **A**, that binds to a protein found in both species of *Plasmodium*. This antibody has a coloured dye attached.
- A test strip is then put into the mixture. The mixture moves up the test strip by capillary action to an absorbent pad.
- Three other antibodies, **B**, **C** and **D** are attached to the test strip. The position of these antibodies and what they bind to is shown in **Figure 7**.



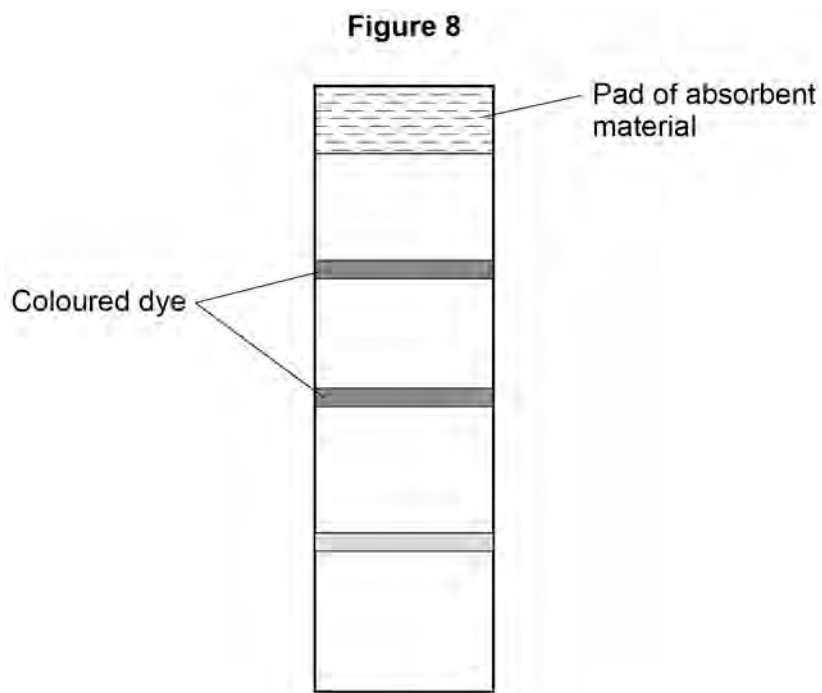
0 8 . 1 Explain why antibody **A** attaches only to the protein found in species of *Plasmodium*.
[2 marks]

0 8 . 2 Antibody **B** is important if this test shows a person is not infected with *Plasmodium*.
Explain why antibody **B** is important.

[2 marks]

Question 8 continues on the next page

- 0 8** . **3** One of these test strips was used to test a sample from a person thought to be infected with *Plasmodium*. **Figure 8** shows the result.



What can you conclude from this result?

Explain how you reached your conclusion.

[4 marks]

[Extra space]
